

**Kolhapur Institute of Technology's
Institute of Management Education and Research (Autonomous), Kolhapur.**

**MCA Department.
MCA-II (Sem-III)-2026-27.**

Mini-Project Synopsis

Date: 03 / 09 / 2026

Roll Number:- MCA2-2026-042	Division:- A
Name of Student: Ms. Ashwini Sawant	
Partner Roll Number(Group Project): N/A (Individual Project)	Division: N/A
Name of Partner(Group Project): N/A	
Project Title: Web-based Faculty Portfolio & Academic Content Management System (CMS)	
Name of Organization & Address (For Live Project):- In-House Academic Project, KIT's Institute of Management Education and Research (Autonomous), Kolhapur.	
Project Guide: Prof. Dr. A. B. Patil / MCA Department Faculty Guide	
Project Description: a. Introduction: The Web-based Faculty Portfolio & Academic Content Management System (CMS) is a modern, full-stack web platform designed specifically for higher education institutions to bridge the gap between academic resource distribution, research dissemination, and student engagement. The platform serves as a unified digital portfolio and CMS for Prof. Ashwini Sawant, enabling students to access course materials, downloadable lecture notes, slide decks, video tutorials, question banks, and published research, while providing faculty with a secure Admin Dashboard to manage and update content dynamically. b. Objectives: - Centralized Digital Profile: To construct a unified online portfolio showcasing academic credentials, teaching experience, research interests, and committee roles. - Resource Distribution Portal: To deliver a structured Teaching Portal where students can search, filter, and download subject-wise notes, PPTs, video links, and question banks. - Research & Publication Repository: To systematically catalog PhD research, peer-reviewed journal papers, and conference proceedings with DOI links and downloadable PDFs. - Student Engagement Analytics: To implement real-time engagement and view duration tracking on academic blogs using browser visibility APIs. - Secure Content Management (CMS): To equip faculty with an intuitive Admin Dashboard for complete CRUD management of profile content, resources, and contact inquiries. c. Problem Statement: In traditional academic setups, educational resources and research records are distributed across fragmented channels (social media groups, cloud drives, email attachments, physical notice boards). This creates key operational challenges: 1. Inconvenient Student Access: Students frequently miss critical lecture notes, model answers, and academic updates. 2. Unstructured Research Record: Faculty publications, conference talks, and certifications are poorly indexed and difficult for external researchers to discover.	

3. Technical Overhead: Traditional websites require manual developer intervention to publish new updates or documents.
4. Lack of Engagement Insight: Faculty members cannot measure student interaction or reading metrics for shared academic content.

d. Proposed Solution:

The proposed system delivers a full-stack Next.js 14 web platform integrated with Prisma ORM and MySQL/SQLite database. It comprises two main operational modules:

1. Public Student & Peer Portal: A responsive user interface containing Home Profile, Teaching Resources (Notes, PPTs, Videos, Question Banks), Research Publications Hub, FDP/Workshops timeline, Certificate lightbox gallery, Analytics Blog, and Interactive Contact form.
2. Admin CMS Dashboard: A secure, authenticated control center enabling full CRUD control over profile details, academic subjects, resources, research records, certificates, blog posts, and contact inbox messages.

e. Front, Back End and Database :

- **Front End:** Next.js 14 (App Router), React 18, TypeScript, Tailwind CSS, Lucide React Icons, Framer Motion.
- **Back End:** Node.js environment, Next.js API Routes (RESTful API), Server Actions, Prisma Client ORM.
- **Database:** MySQL Database / SQLite (configured via Prisma ORM with relational schema integrity).

f. System Requirements(H/w-S/w):

1. Hardware Requirements:

- **Processor:** Intel Core i3 / i5 / AMD Ryzen 5 or higher
- **System Memory (RAM):** 8 GB minimum
- **Storage Space:** 20 GB available SSD space
- **Display Resolution:** 1920 x 1080 (Full HD)
- **Internet Connection:** High-speed broadband network connection

2. Software Requirements:

- **Operating System:** Windows 10 / 11 (64-bit), Linux, or macOS
- **Environment Runtime:** Node.js v18.x or v20.x LTS
- **Database Engine:** MySQL 8.0+ / SQLite 3
- **Core Framework:** Next.js 14 (React 18, TypeScript 5)
- **Package Manager:** npm / yarn / pnpm
- **Web Browser:** Google Chrome, Mozilla Firefox, Microsoft Edge, Safari